

Information Retrieval Project

Module 1: Web Crawler

Crawler Requirements

1. The crawler should download **at least 3000** documents from the website: **www.cnn.com** or **www.bbc.com** or **www.wikipedia.com**
2. You can use any DBMS **to store the page content**:
3. **ONLY English Web pages** are stored. Ignore pages with other languages

Module 2: Indexing

1. Parses HTML pages to extract text.
2. Tokenizes text using spaces and commas.
3. Applies preprocessing: punctuation removal, case folding, stop word removal, and stemming(such as Porter Stemmer).
4. Builds an inverted index with term, doc_id, frequency, and position like this example:

Term	Doc_id	Frequency	Positions
search	1	2	[3, 15]

Module 3: Ranking (Scoring)

1. Processes user query using same preprocessing steps as indexing.
2. Retrieves term statistics from the inverted index.
3. Computes TF, IDF, and TF-IDF weights.
4. Calculates score for each document based on query terms.
5. Sorts documents in descending order of relevance score.

Module 4: Retrieval

1. Accepts user query and preprocesses it.

2. Searches the inverted index to retrieve matching documents.
3. Supports normal and wildcard queries (e.g., comput*, *ing, te?t).
4. Displays ranked results (document ID/URL and score):

Query: data mining: the output will be

Rank	Document ID	Score
1	Document 12	0.92
2	Document 7	0.85
3	Document 3	0.80

All the best, with my best wishes.